

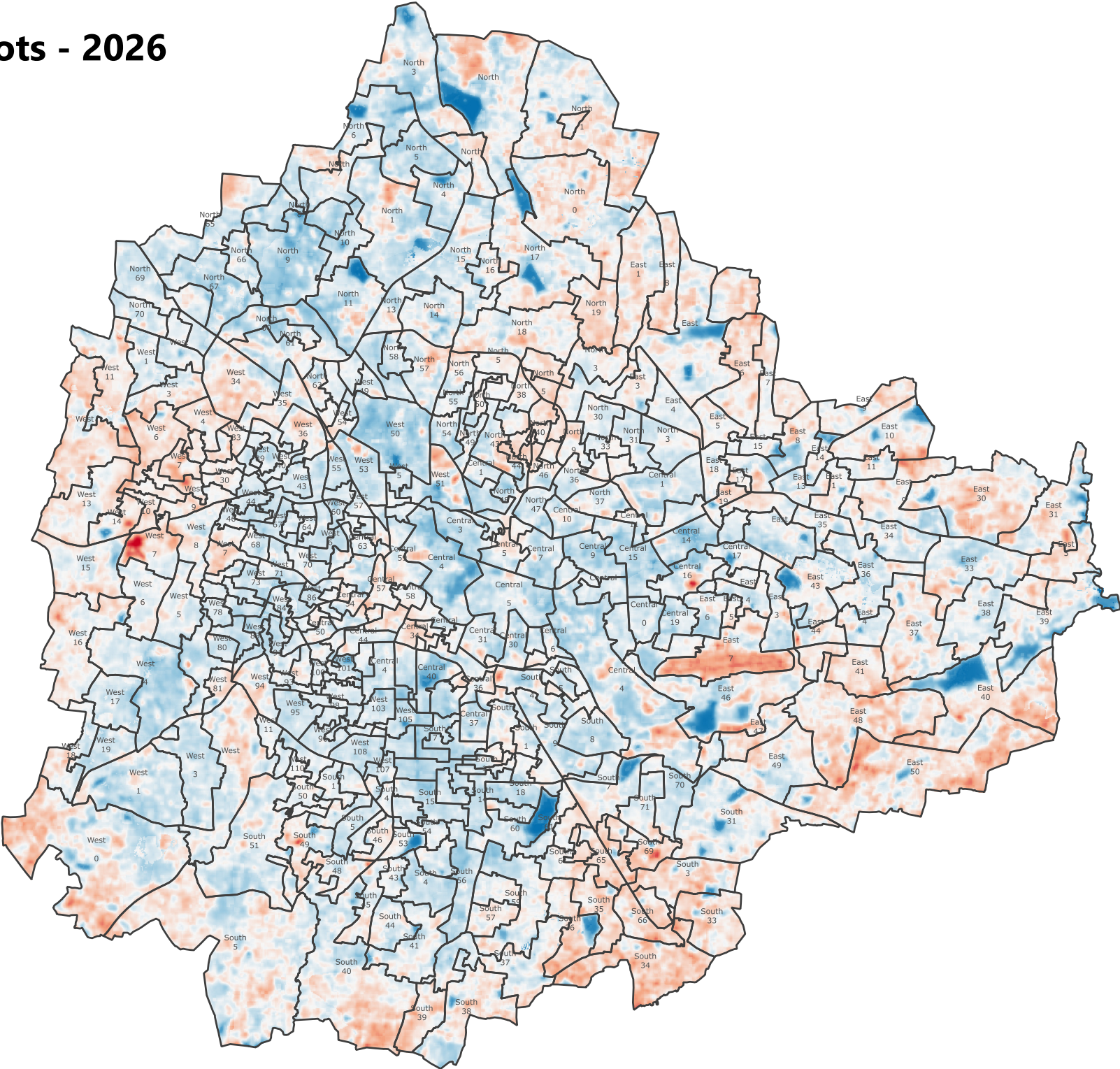
Urban Heat Hotspots - 2026

Legend:

Urban Heat Hotspots - 2026



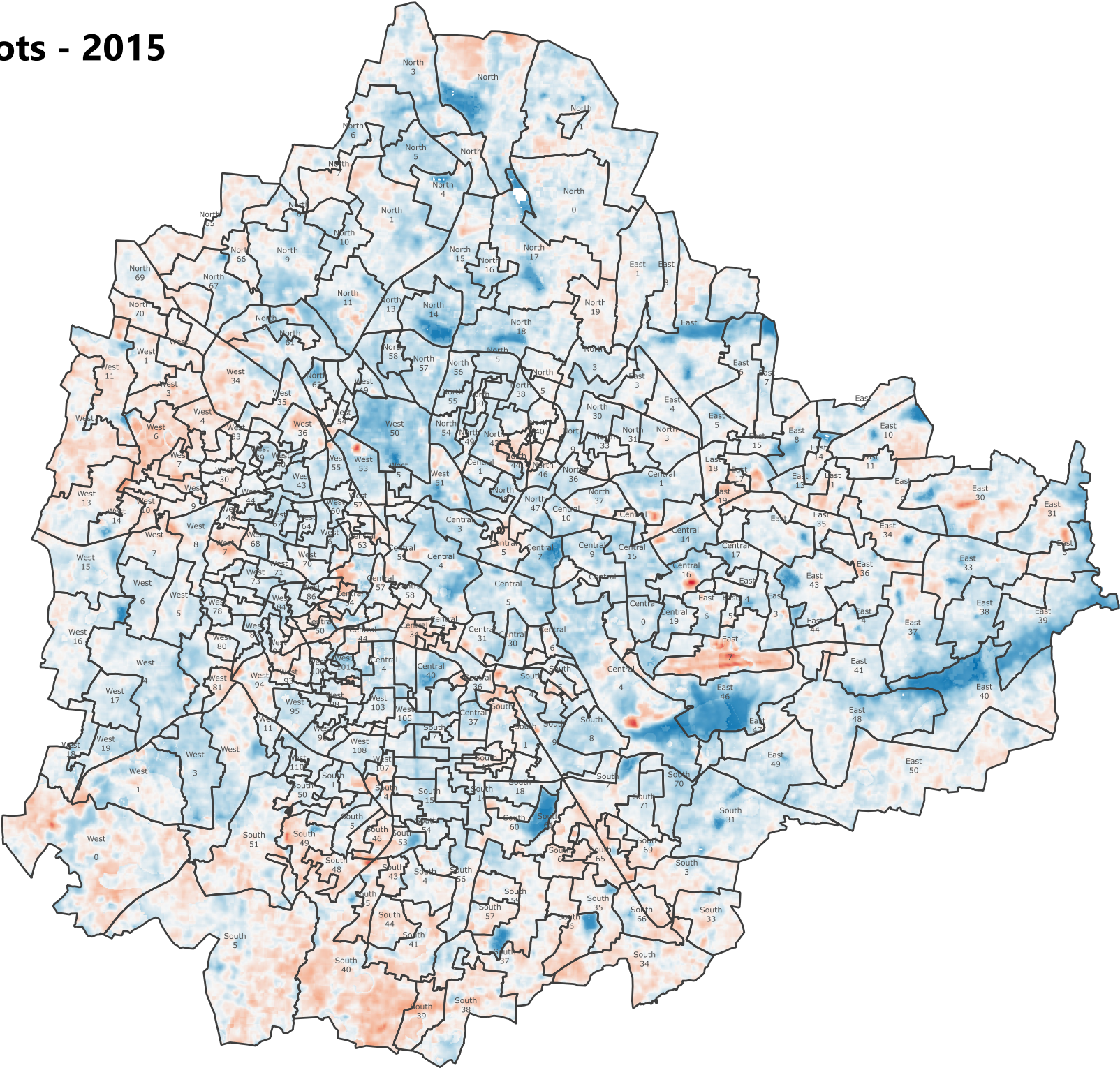
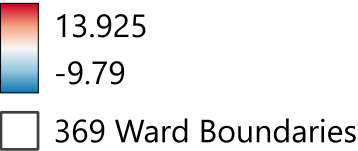
369 Ward Boundaries



Urban Heat Hotspots - 2015

Legend:

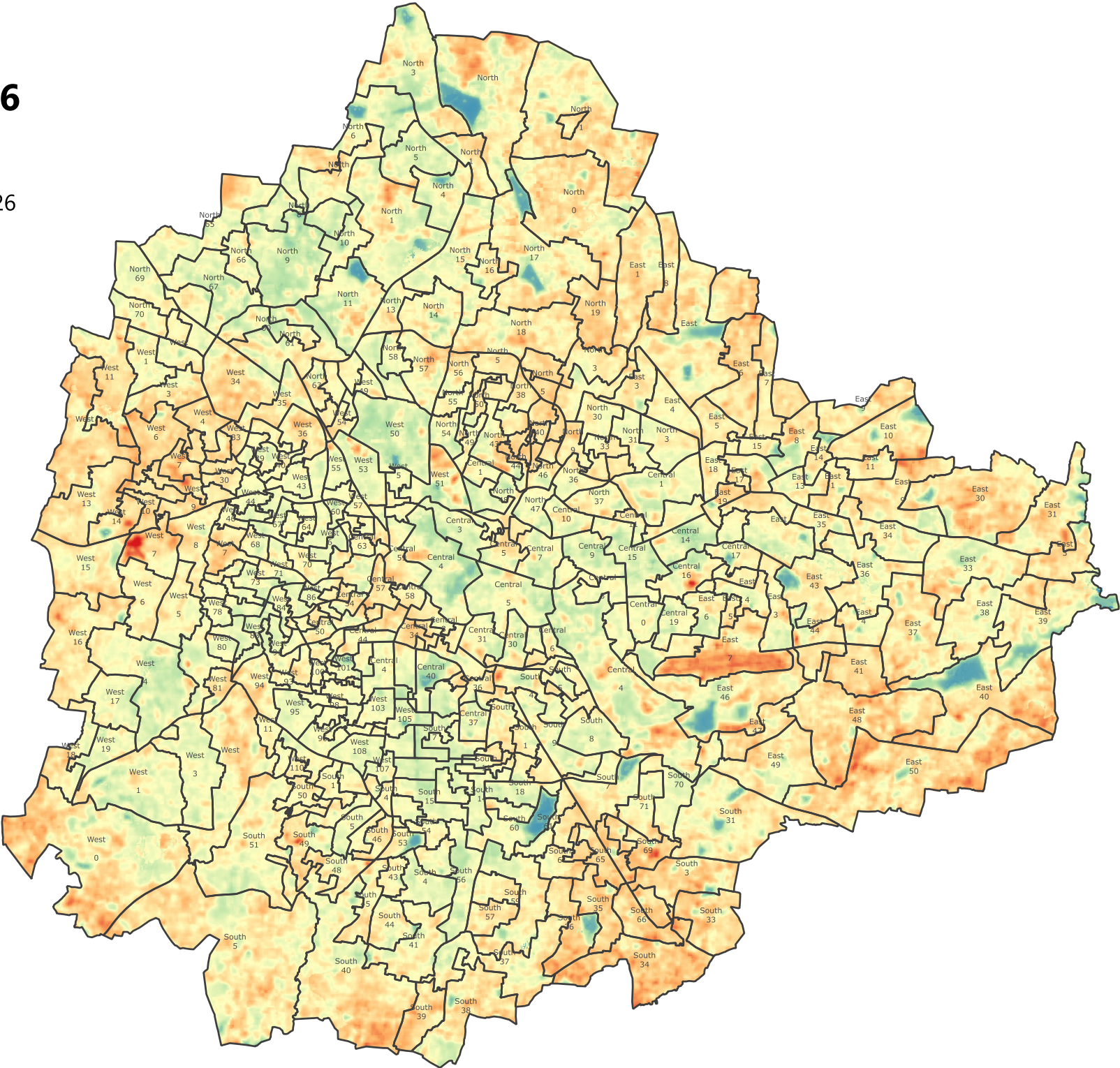
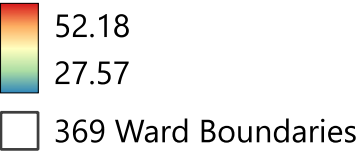
Urban Heat Hotspots - 2015



Land Surface Temperature - 2026

Legend:

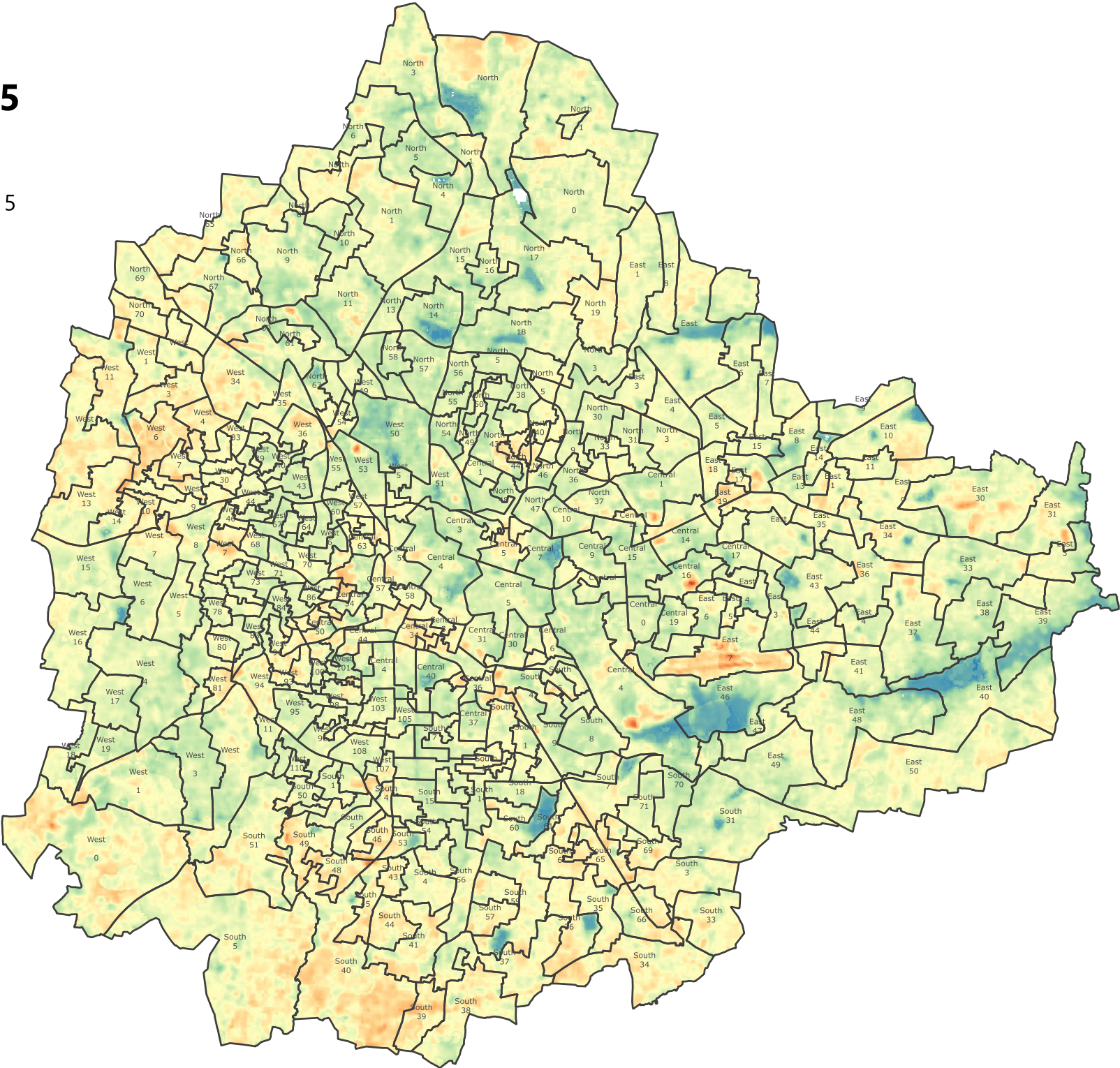
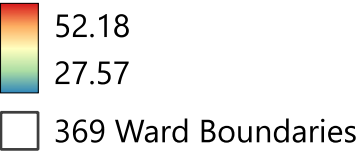
Land Surface Temperature - 2026



Land Surface Temperature - 2015

Legend:

Land Surface Temperature - 2015



Methodology

Two variables were mapped across the city's wards:

1. Land Surface Temperature
2. Urban Heat Hotspots

Land Surface Temperature (LST) is the temperature of the ground i.e. rooftops, roads, soil, parks, and water, as measured from space. It is not the air temperature that is felt, but the heat radiated from surfaces, which is what drives a neighbourhood's warmth. It was measured in degrees Celsius.

Urban Heat Hotspots: A single reference temperature was first calculated: the average surface temperature of all natural, unpaved (permeable) surfaces across the entire study area. The UHI value at each location was then found by subtracting this one city-wide reference from that location's own temperature. A positive value therefore indicates a place hotter than the city's overall permeable average, for example, +5 °C means 5 °C above that reference, while a negative value indicates a cooler place, typically a park, lake, or water body. Zero marks a location sitting exactly at the reference temperature.

Data sources

- **Ground temperature:** Landsat 8 and 9 (NASA / U.S. Geological Survey). Land surface temperature was derived from the Landsat 8 and 9 Collection 2 Level 2 surface temperature product. Landsat is a joint NASA–USGS programme of Earth-observing satellites that has imaged the planet continuously since 1972; the two current satellites together revisit any location roughly every eight days. These Level 2 products are supplied already atmospherically corrected and radiometrically calibrated by the USGS and are internationally recognised as Analysis Ready Data. The temperature is obtained from the satellites' thermal infrared band; although it is delivered by the USGS resampled to a 30-metre grid, the true resolution of the thermal sensor is about 100 meters.
- **Land cover:** Google Dynamic World. Built-up land was separated from natural, permeable surfaces using Google Dynamic World. This is a near-real-time global land-cover map produced automatically by a deep-learning model applied to 10-metre Sentinel-2 satellite imagery, updated every two to five days and classifying the surface into nine categories such as trees, grass, water, and built-up area. It was used only to identify which pixels are built-up (impervious) and which are permeable (all other classes).
- **Processing platform: Google Earth Engine.** All satellite data were accessed and processed in Google Earth Engine (Gorelick et al., 2017), a free, cloud-based platform for planetary-scale geospatial analysis. It hosts the full Landsat and Dynamic World archives and performs the computation on Google's servers, which removes the need for large volumes of imagery to be downloaded and processed locally.

Analyses

All processing was performed in Google Earth Engine. For each year, every clear satellite image from the hot pre-monsoon season (mid-March to mid-May, when heat stress peaks) was combined into a single

average temperature map, with any pixels obscured by cloud or cloud shadow automatically discarded. From these, the heat island effect was calculated for every location by subtracting one fixed reference, the average temperature of all permeable surfaces across the whole study area, from each pixel's temperature. The results were then averaged within each ward so that they could be compared fairly across the city and between years.

How to read the maps:

On temperature maps, blues mark cooler ground and reds mark hotter ground. On heat island maps, blue areas are cooler than the city's natural baseline and red areas are hotter, with the strongest reds marking the most intense heat build-up.

Limitations

- Temperatures are measured from the ground surface, not the air, so the values reflect surface heat, which is generally more extreme than the air temperature people actually feel.
- The satellites pass over at a fixed mid-morning time (around 10:30 a.m.), so the maps capture that moment only — not midday, evening, or overnight conditions.
- The thermal sensor's true resolution is about 100 metres, so each value is an average over roughly a city block; fine features like a single street or narrow park cannot be resolved.
- Satellites cannot see the ground through cloud. Cloudy pixels were removed automatically, but cloudy periods leave fewer valid observations and undetected haze can bias individual pixels.
- A satellite revisits only about every 16 days, so a season yields few passes and the average rests on a small sample. This is more pronounced for 2015, when only Landsat 8 was in orbit (Landsat 9 launched in late 2021), giving earlier years fewer observations than recent ones.
- Satellite thermal readings depend on assumptions about how efficiently each surface radiates heat, which introduces some uncertainty into absolute values, though comparisons between similar surfaces remain reliable.

References

Earth Resources Observation and Science (EROS) Center. (2020). Landsat 8–9 Operational Land Imager / Thermal Infrared Sensor Level 2, Collection 2 [dataset]. U.S. Geological Survey.
<https://doi.org/10.5066/P9OGBGM6>

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Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202, 18–27.

A short credit line is also requested by the USGS for Landsat products, which may be added to any figure: "Landsat Level-2 Surface Temperature Science Products courtesy of the U.S. Geological Survey."